

Session 20: AI Bias & Fairness

You've seen AI doing amazing things. But sometimes AI gets things wrong in **unfair ways**. Today, we find out why - and what to do about it. This is one of the most important sessions in the course. ⚖️



Objectives

By the end of this session, you'll be able to:

01

Define AI Bias

In your own words

03

Spot 3 Sources

Where bias sneaks into AI

05

Detect & Reduce

Understand how to fight bias

02

Confidence vs. Fairness

Tell the difference

04

Real Stories

Hear where AI was unfair

06

Your Role

Know how YOU can make AI more fair



Recap & Homework

Show your improved Lovable app 🚀

- What did you change?
- Did you start a second app?

Quick check from last Session

- AI is changing jobs, not always replacing them.
- What are the 5 skills that matter?

Warm-Up

Has This Ever Happened to YOU?

Two kids did the same thing - but only ONE got in trouble?

A team picked players - and you got skipped for an unfair reason?

Someone made a quick judgment about you before they really knew you?

How did that feel?

Here's the thing: unfairness like this usually happens when someone makes a decision based on incomplete or biased information. AI can do this too but AI doesn't *mean* to be unfair. It doesn't have intentions or feelings. It just finds patterns in data and if the data has unfair patterns in it, AI learns those too.

That's what we're digging into today.

What Is AI Bias?

AI Bias is when AI works better for some people than others. Not because it *wants* to - AI has no feelings. But because it learned from unfair or incomplete data.

Systematically Wrong

The AI is consistently wrong for one specific group of people.

Repeating Pattern

It happens over and over - not just by accident once.

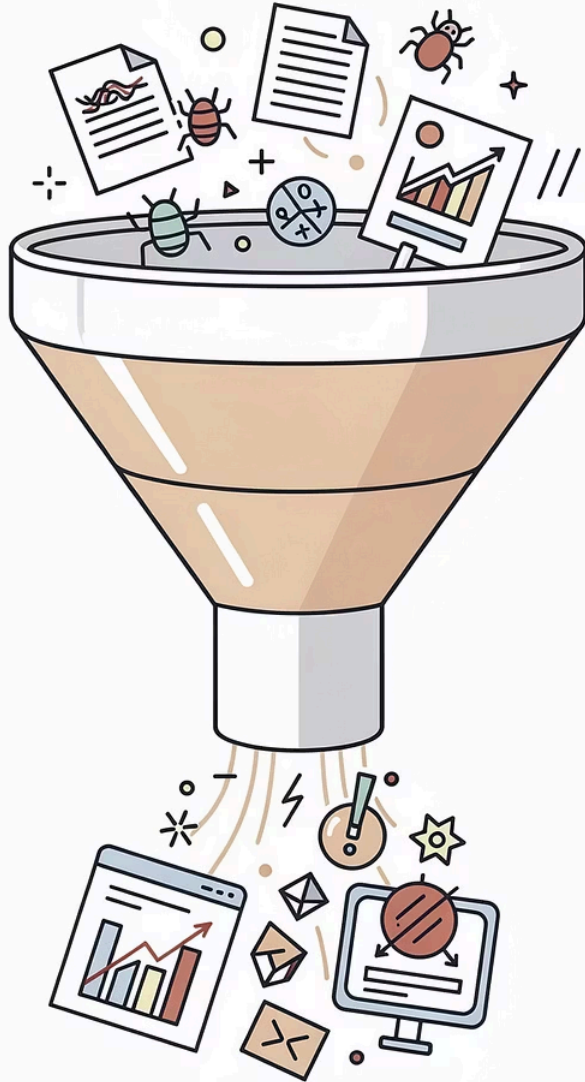
Real Harm

It can and does hurt real people in real situations.

⚠ Bias isn't a glitch. It's a **pattern**.



Garbage In



Garbage Out

AI Doesn't MEAN to Be Biased

AI isn't *choosing* to be unfair

AI doesn't have feelings

AI doesn't have opinions

AI doesn't know what
fairness is

AI is a **pattern-finder**. And if its data is biased... then its patterns will be biased.



Remember "Garbage In, Garbage Out" This is exactly that but with real-world consequences for real people.

Where Does Bias Come From?

Bias sneaks in through **3 doors**. We'll look at each one.



DATA BIAS

The training data doesn't include everyone fairly.



ALGORITHM BIAS

The AI's "recipe" itself causes unfair patterns.



HUMAN BIAS

The people who BUILT the AI brought their own biases with them.

DATA BIAS

The training data isn't fair to everyone.

Example: Imagine training a voice assistant - but most of the training audio is American-accented adults.

✓ Works GREAT for:

- American-accented adults
- Standard speech patterns
- Adult vocal ranges

✗ Struggles for:

- Kids' voices
- Indian accents
- Scottish accents
- Anyone with a speech difference

 **Why?** Because AI only knows what it was shown.



ALGORITHM BIAS

Sometimes the algorithm itself **amplifies unfairness**.

YouTube's algorithm has one goal: **keep you watching**. So it shows you whatever keeps people watching longest - no matter what that content actually is.

What the Algorithm Discovered

 Angry videos hook people more effectively than calm ones

 So the algorithm shows **more** of that content even though no one told it to

Was the Algorithm Trying to Be Unfair?

The algorithm wasn't trying to be unfair. It had no malicious design, no biased data, and no bad actors behind it.

The Real Problem

Maximizing watch time isn't the same as showing GOOD content.



HUMAN BIAS

People who build AI bring their own biases - **often without knowing it.**

Unconscious Assumption

A team building a hiring AI might unconsciously think "great engineers look like THIS."

Biased Labeling

They label training data based on that assumption and pick features that reflect their idea of "good."

AI Mirrors the Bias

The AI mirrors their biases - at **huge scale**. That's why diverse teams build better AI.



More viewpoints = fewer blind spots.

Mini Activity 1 - Spot the Source!

For each problem below, which source of bias caused it? **Data / Algorithm / Human?**

#	Problem	Data / Algorithm / Human?
1	Face recognition trained on mostly light skin - fails for dark skin	?
2	YouTube recommends extreme videos because they get clicks	?
3	Hiring AI prefers names that "sound American" because the team only saw those work before	?
4	Voice assistant trained mostly on American English - struggles with Indian accents	?
5	Social media algorithm keeps showing kids unhealthy body content because it gets engagement	?

 Think about each case: Was it the **data**, the **algorithm's goal**, or the **people who built it**?

Mini Activity 1 - Answers

1

DATA Bias

Face recognition fails for dark skin - training data was not diverse enough.

2

ALGORITHM Bias

YouTube recommends extreme videos - optimized for watch time, not quality.

3

HUMAN Bias

Hiring AI prefers some names - team's assumptions baked into the model.

4

DATA Bias

Voice assistant struggles with accents - not enough diverse audio in training.

5

ALGORITHM Bias

Algorithm shows unhealthy content - engagement metric rewarded harmful material.

☑ See the pattern? Bias has a **SOURCE**. Find the source → you can start to fix it.

Gender Shades (2018)

A researcher named **Joy Buolamwini** at MIT tested 3 major face-recognition AI tools.

✅ Light-skinned men

Mistakes less than **1%** of the time.

❌ Dark-skinned women

Mistakes **34%** of the time!

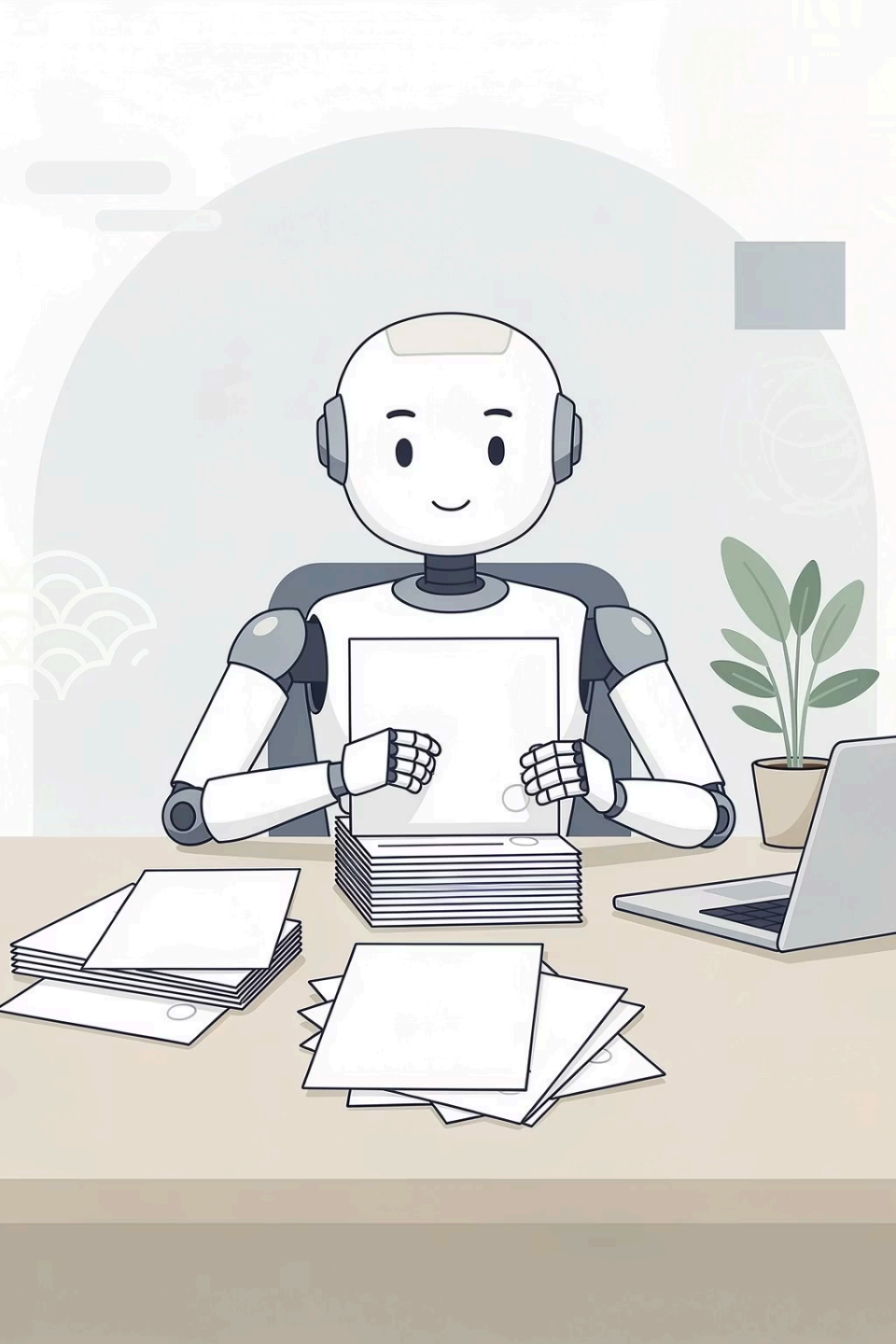
🤔 Why?

Training data was mostly light-skinned faces - not representative of everyone.

Result: Companies like IBM and Microsoft fixed their models. Joy started a movement called the "**Algorithmic Justice League**" - real people fighting AI bias. ⚖️

✅ AI bias can be **measured** AND it can be **fixed**.





Amazon's Hiring AI (2018)

What Happened

Amazon built an AI to help hire engineers. It trained on **10 years of past hiring data**.

🤔 The problem? The past hiring data had **mostly men**. So the AI learned: *"Men = good hires."*

It started **lowering scores** for resumes that said "women's" - like "women's chess club" or "women's college."

The Outcome

🛡️ **Good:** Amazon caught it and SHUT THE PROJECT DOWN before deploying.

⚠️ **Reality check:** How many other AIs are doing this RIGHT NOW and we don't know?

COMPAS (Criminal Justice - 2016)

US courts started using an AI tool called **COMPAS** to predict if someone might commit another crime.

This affected **REAL decisions** - who got bail, who got parole, who stayed in jail.

Investigators found: COMPAS labeled Black defendants as "high risk" much more often than White defendants - even when their actual crime histories were similar.

- ⊗ This is what we mean by "real consequences." When AI affects people's lives, we can't just trust it. We have to **AUDIT** it.



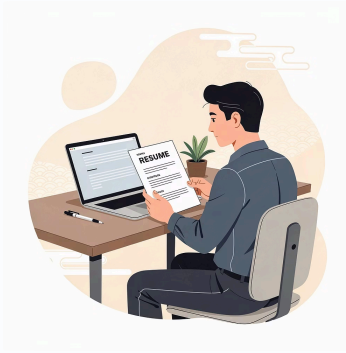
Why This MATTERS

AI bias isn't just a "tech problem." It affects real people in every area of life.



Healthcare

Who gets medical treatment and accurate diagnoses.



Employment

Who gets hired, promoted, or passed over.



Justice

Who gets sent to jail or denied parole.



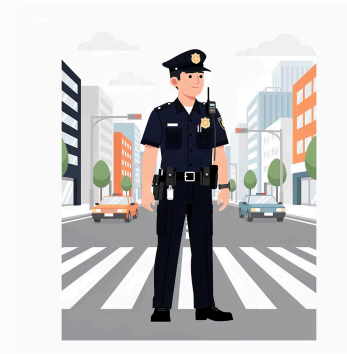
Housing & Loans

Who gets a mortgage or financial opportunity.



Education

Who gets into schools and receives opportunities.



Policing

Who gets stopped, searched, or flagged by authorities.

⊗ Real people. Real lives. Real harm. That's why this matters more than almost any other AI topic.

How Do We DETECT Bias?

You can't fix what you can't see. So we have to **look** for bias.

Test on Different Groups

Does the AI work equally well for everyone? Run it on diverse populations and compare results.

Check the Training Data

Was every group represented fairly? Look at who's in the data - and who's missing.

Audit the Outcomes

Are some groups getting systematically different results? Track patterns over time.

- ☑ This work is called an **AI Audit**. It's a real job. 🛡 Remember the Guardians? This is what they do - every day, in real companies, all around the world.

How Do We REDUCE Bias?

Once we find bias, we can fight it. Real ways to make AI more fair:



Better Data

Make sure all groups are represented in training data - not just the majority.



Diverse Teams

Different people catch different blind spots. Diversity in the room = diversity in the model.



Test, Test, Test

Before AND after launching. Bias can appear at any stage.



Human-in-the-Loop

Humans review AI decisions that affect people's lives - don't let AI decide alone.



Transparency

Companies share what their AI does and how - so the public can hold them accountable.

This is hard work. But it's how we make AI worth trusting.

Mini Activity 2 - Be an AI Auditor!

Pretend you're an AI auditor. For each AI, ask: **WHO** might it be unfair to?

AI System	Who might it be unfair to?
 Face Unlock on phones	?
 Voice assistants like Alexa	?
 An essay-grading AI for schools	?
 An AI that predicts medical risk	?
 An AI that recommends college majors	?

 Use what you know: Think about who's likely **in** the training data - and who's **NOT**.

Mini Activity 2 - Possible Answers

There's no single "right" answer. Here are some possibilities:

AI System	Who might it be unfair to?
 Face Unlock	Dark-skinned people, kids with smaller faces, people with disabilities
 Voice Assistants	Non-American accents, kids' voices, people with speech differences
 Essay Grader	Non-native English speakers, students who write differently
 Medical AI	Groups not well-represented in medical research
 College AI	Kids from less-known schools, kids with non-traditional paths

🕒 See the pattern? Whoever isn't well-represented in the training data → **that's who AI fails.**

YOUR Role in Fair AI

You don't have to be an engineer to help. Here's what you can do **right now**:

Ask Questions

When AI gives unfair-feeling results, speak up and ask why.

Speak Up

When you see something wrong, say something. Don't stay silent.

Test AI

Try it on yourself, on different people you know. See what you find.

Share Stories

Share stories of AI failures so they get noticed and fixed.

Build It Better

Use your AI literacy to make fair tools when you create.

 You're already asking questions. That's the first step. 

Hands-On Activity - Be a Junior AI Ethicist

Time to investigate a real bias case - the way professional AI ethicists do.

The Case: "Photo Tagging Gone Wrong"

A social media company built an AI to auto-tag photos with labels like "beach day" or "birthday party."

What Users Noticed

Photos of people with darker skin tones were mislabeled or got **no labels at all**. Searching "outdoor activity" mostly returned lighter-skinned people.

What Engineers Found

The training data had **far fewer photos of diverse skin tones**. Classic data bias.

The Impact

The company paused the feature. **Users felt invisible**. Your mission: figure out what went wrong and propose a fix.



Hands-On - Your Junior Ethicist Brief



Step 1: Open NotebookLM

Go to notebooklm.google.com → create a new notebook → paste the case study from the previous slide as a source.



Step 2: Ask 4 Questions

1. What was the AI feature designed to do? 2. What evidence of bias did users report? 3. What did engineers find about the training data? 4. What was the real-world impact on users?



Step 3: Propose 3 Fixes

 Fix #1: _____

 Fix #2: _____

 Fix #3: _____

What We Learned in this Session

✓ AI Bias Defined

When AI works systematically better for some groups than others - not by choice, but by pattern.

✓ 3 Sources of Bias

Data (unfair training data), **Algorithm** (wrong optimization), **Human** (builders' blind spots).

✓ Real-World Stories

Gender Shades · Amazon Hiring AI · COMPAS - bias with real consequences for real people.

✓ Detection Methods

Test on different groups, check training data, audit outcomes. This is a real job.

✓ Reduction Strategies

Better data, diverse teams, testing, transparency, Human-in-the-Loop.

✓ Your Role

Ask questions, speak up, test AI, share stories, build fair tools.

Your Homework - Junior AI Ethicist Report

Spot the Bias

Pick **2 AI tools** you use. Test each one. Look for bias.

Ideas to Test:



Face Unlock

Try with different family members



Voice Typing

Try with different accents



ChatGPT

Ask about people from different countries/cultures



Image Generator

Ask for "a doctor" or "a CEO" - what does it show?

For EACH tool, fill in:



Tool Name

Identify the AI tool you are evaluating — be specific about its function and context of use.



Bias Spotted

Describe the bias you observed. What patterns, outputs, or behaviors raised a red flag?



Who Gets Hurt

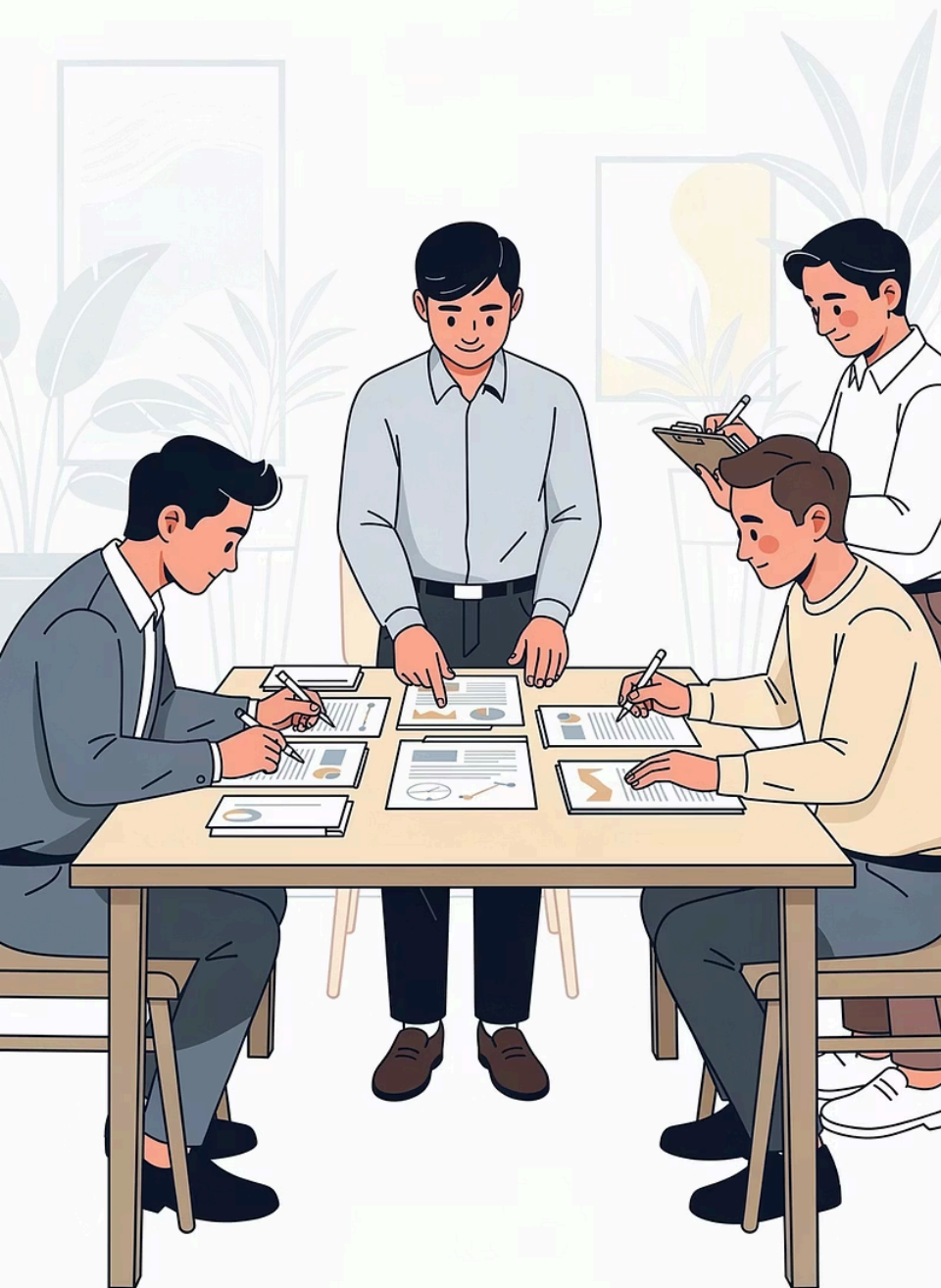
Name the specific group or community that could be harmed by this bias in real-world use.



One Concrete Fix

What is one actionable change the company could make to reduce or eliminate this bias?

Finish this sentence: *"The most important thing I learned about AI fairness this week is..."*



Next Time - Session 21:

AI Safety, Misuse & Governance

You've seen how AI can be **unfair**. Next session: how can AI be **dangerous**?



Worst Case Scenarios

What's the worst that could happen with AI? We'll explore real risks.



Deepfakes

What are deepfakes - and why are they scary? How do you spot one?



AI Governance

How do we govern AI to keep people safe? Who's in charge?



Rules for AI

What rules do we need? Who makes them? How do they get enforced?